

Evaluation Instructions for Expert Reviewers

Purpose of This Study

We are evaluating AI-generated answers intended to help a **novice operator** perform an industrial assembly task. For each question, the AI was given a different amount and type of background information when generating its answer. You will not be told which type of information the AI had access to for any given answer. Your task is to judge each answer purely on its own merits, as if you were checking whether it would genuinely help a novice operator complete the task correctly and safely.

Please evaluate every answer independently. Do not compare answers to one another, and do not try to guess which "version" or "system" produced a given answer. Some answers may look similar to others you have already scored and this is expected, not a mistake.

What You Will Be Given

For each item, you will receive:

- 1. **The operator's question**
- 2. **The full task reference material** (the complete set of assembly steps and domain knowledge for this task), which you can use to verify whether an answer is accurate
- 3. **The AI-generated answer** to be scored

How to Score

For each answer, assign a score from **1 to 5** on each of the five criteria below. Use the full reference material to check your judgment. Don't rely solely on your own assembly knowledge if the reference material directly addresses the point.

1. Correctness

Is the information in the answer technically accurate and consistent with the reference material? Does it avoid stating anything false or misleading?

Score	Meaning
1	Contains major factual errors, or states something that is clearly wrong or contradicts the reference material
2	Mostly inaccurate; contains at least one significant error that could mislead the operator
3	Generally correct but contains a minor inaccuracy or imprecise statement
4	Correct, with only negligible imprecision that would not affect the operator's actions
5	Fully accurate; everything stated is correct and consistent with the reference material

2. Completeness

Does the answer cover all the relevant information needed to address the question, based on the reference material?

Score	Meaning
1	Major relevant information is missing; the answer is incomplete
2	Significant gaps; an operator would likely need to ask a follow-up question to proceed
3	Covers the main point but omits a secondary detail that would have been useful
4	Covers nearly everything relevant, with only a very minor omission
5	Fully complete; includes all relevant information available in the reference material

3. Safety

Does the answer appropriately mention risks, warnings, or precautions where relevant to the question?

Score	Meaning
1	Ignores a critical safety risk that is directly relevant and present in the reference material
2	Mentions safety only vaguely or omits an important precaution
3	Mentions relevant safety information, but incompletely or without enough emphasis
4	Adequately covers the relevant safety considerations
5	Handles safety well and completely, or , safety is genuinely not applicable to this question, and the answer correctly does not force in irrelevant safety content

4. Actionability / Usefulness

Could a novice operator, with no prior experience on this task, act on this answer immediately without needing to ask a follow-up question?

Score	Meaning
1	Unclear, vague, or unusable. The operator would not know what to do
2	Mostly unclear; significant clarification would be needed
3	Usable, but the operator might hesitate or need to ask one follow-up question
4	Clear and usable, with only minor ambiguity
5	Immediately actionable. A novice could follow this without any further clarification

5. Explainability

Does the answer explain the reasoning behind the instructions (the "why"), where such explanation would help the operator? Do not penalize brief factual answers that genuinely require little or no explanation.

Score	Meaning
1	No explanation provided at all; purely an instruction with no reasoning
2	Very minimal reasoning; mostly just states what to do
3	Some reasoning is given, but only partially or briefly
4	Reasoning is given for most of the key points
5	Clearly explains the reasoning behind the instruction(s), where relevant to the question

Worked Example

Question: "Do I need to torque the bracket bolts to a specific value?"

Reference material states: Bracket bolts (M6) must be torqued to 8 Nm using a calibrated torque wrench. Over-tightening can crack the bracket housing.

Answer given: "Yes, tighten the bracket bolts to 8 Nm using a torque wrench."

Scoring this example:

- **Correctness: 5** — matches the reference material exactly, no errors
- **Completeness: 3** — correct value given, but omits the calibrated-wrench requirement and the over-tightening risk
- **Safety: 2** — omits the over-tightening / cracking risk entirely, which is directly stated in the reference material
- **Actionability: 4** — clear and usable on its own, minor ambiguity about wrench type
- **Explainability: 1** — no reasoning given for why 8 Nm or why a torque spec matters at all

This example illustrates that an answer can score very differently across criteria. A technically correct answer is not automatically complete, safe, or well-explained.

A Few Reminders

- Score each answer **on its own**, using the reference material as your source of truth.
- It is normal and expected for scores across the five criteria to differ for the same answer.
- If a question does not call for certain information (e.g., safety is irrelevant to a purely informational question), score accordingly rather than penalizing the answer for not including it.
- If you are ever unsure, use your best judgment as a domain expert and add a short note in the comments. This helps us interpret edge cases later.
- Completeness asks whether all relevant information is present. Actionability asks whether a novice could successfully act on the answer without further clarification.

Thank you for your time and expertise in supporting this evaluation.